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RADIATION FOG INGREDIENTS

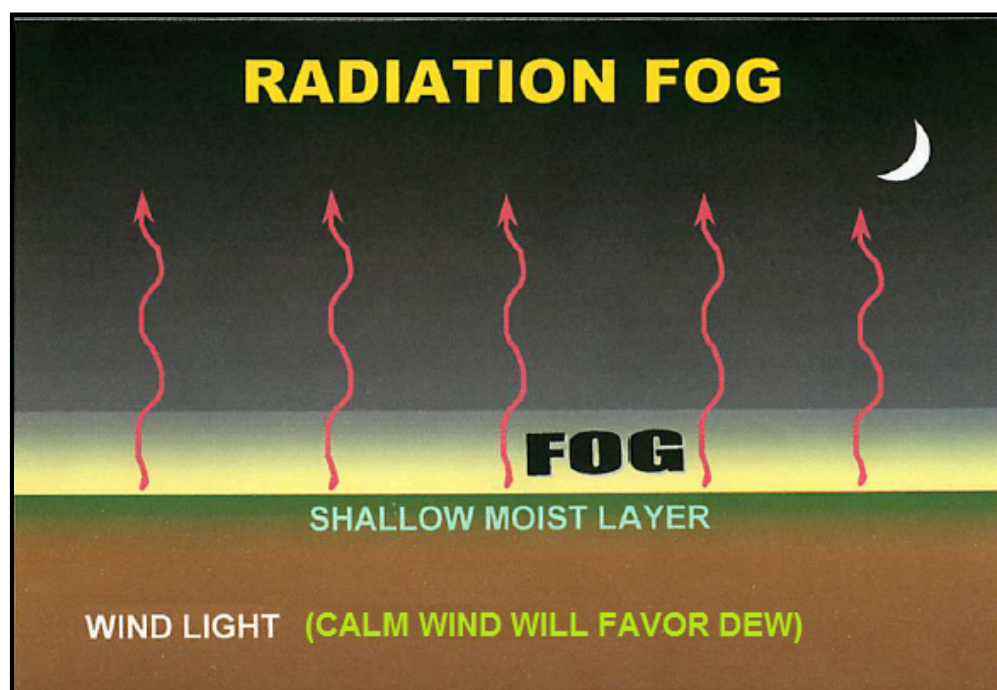
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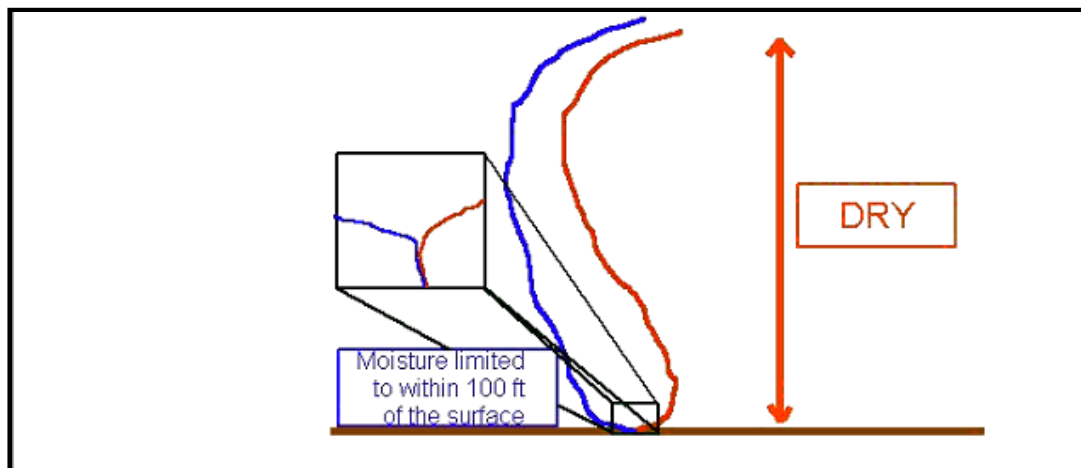
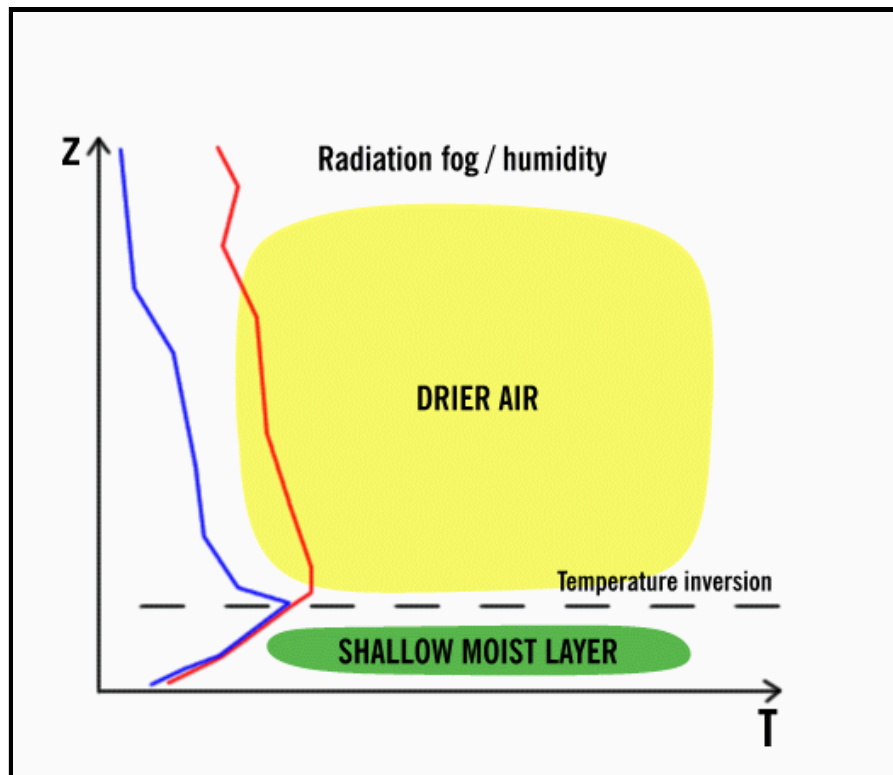
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The prime time ingredients for radiation fog are saturated soil, light wind, initially clear skies, and a low afternoon dewpoint depression. The more factors that are present, the more likely the fog will be. Saturated soils continuously evaporate moisture into the air, insuring the dewpoint depression (difference between temperature and dewpoint) will remain low. Light wind reduces the amount of mixing of air in the PBL. If winds are light, moisture evaporating from the surface will remain near the surface and not mix with drier air aloft. If wind is calm, expect fog to be very close to the ground or non-existent. In a calm situation with a low dewpoint depression and moist soils, expect a thick dew or frost instead of radiation fog.

Clear skies allow the maximum amount of longwave radiation to leave the earth. The absence of clouds will prevent any of the radiation from being trapped between the cloud and the ground. The more temperatures cool, the quicker the temperature will reach the dewpoint. A low dewpoint depression can occur by adding moisture to the air while at the same time cooling the air. The best way to rapidly decrease the dewpoint depression is for it to rain. An afternoon rain increases the likelihood of overnight fog dramatically. The afternoon rain saturates the soil and reduces the afternoon dewpoint depression. If skies begin to clear at night and winds are from 5 and 10 miles per hour after rain occurred the preceding day, fog is extremely likely. Other processes can produce fog such as upslope flow and contact cooling.

Radiation fog will dissipate from daytime heating. This is due to surface air warming by conduction and becoming unsaturated. Radiation fog usually dissipates by mid-morning.





This is the type of sounding that occurs with a purely radiation fog event. This sounding is never found with advection or marine fog and rarely with frontal fog. If the height of the lowest inversion is 100 feet, then that will be the maximum height this fog may reach. Although, radiation fog does not always follow this rule and it may be well beneath this height. It does not move much and develops from ground up. It is never very deep which means it burns off faster. The wind is calm to very light. This type of fog usually takes place as post-rain events up to 36 hrs or moisture increases during the day. When there is not sufficient moisture depth, this fog develops near the ground and stays within 2 feet of the surface. The radiation caused inversion starts almost immediately off the ground to 100 feet on average. Only when enough moisture is present will this fog get to 1/4 of a mile in visibility. Usually this requires some type of moisture advection process from the gulf or warm marsh and lake waters during daylight hours. Clear skies will always be needed as well. Radiation fog rarely envelopes the downtown areas of a city. The heat island effect is normally strong enough to keep radiation fog from forming.

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FORECAST ADVECTION FOG

METEOROLOGIST JEFF HABY

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Advection fog is fog produced when air that is warmer and more moist than the ground surface moves over the ground surface. The term advection means a horizontal movement of air. Unlike radiation fog, advection fog can occur even when it is windy. Also unlike radiation fog, advection fog can occur when the skies aloft are initially cloudy.

The setup for advection fog will often include an advection pattern bringing in warmer and more moist air from the south. The set-up for the ground surface will be a snow covered ground or a saturated ground that has been chilled by cold temperatures before the winds shift back from a southerly type direction.

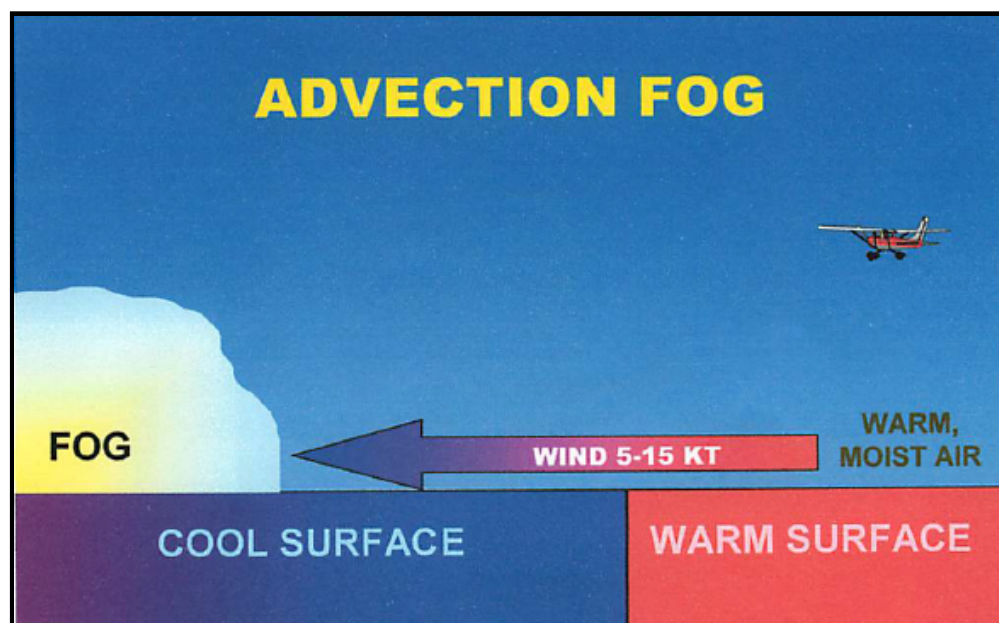
Since the ground surface is very cold it will influence the temperature of the air adjacent to the ground surface. This air will be chilled more than it otherwise would be due to the very cold surface ground temperature. If there is snow or moisture on the ground then the air will be cold and moist. When winds shift to the south it will bring in warmer air. This warmer air will be cooled due to the influence of the cold land surface. As air cools the temperature drops closer to the dewpoint. If the mixing of the warmer air with the colder air produces a relative humidity of 100% then fog can form.

Considering air that is saturated, as the temperature increases the amount of moisture in the air increases at an increasing rate. Warm air that is near saturation will saturate quickly when it is mixed with cold saturated air. This is because the amount of moisture in the air from mixing the air is greater than the amount of moisture needed for saturation at the temperature of the mixed air. The air, instead of supersaturating, will produce condensation in the form of fog.

For example, suppose air that has a temperature and dewpoint of 0 C is mixed with air that has a temperature and dewpoint of 20 C. The temperature of mixing this air if it is mixed in equal proportions is 10 C. A saturation vapor pressure at 10 C is 12.3 mb. The air with a temperature of 0 C has a saturation vapor pressure of 6.1 mb and the air with a temperature of 20 C has a saturation vapor pressure of 23.4 mb. When this air is mixed the new saturation vapor pressure is $(6.1 + 23.4) / 2 = 14.8$ mb. Since 14.8 is the amount of moisture in the air after mixing and mixed air at 10 C only needs 12.3 mb to be saturated, the air is supersaturated. Instead of the Relative Humidity increasing above 100% though, condensation and fog will form.

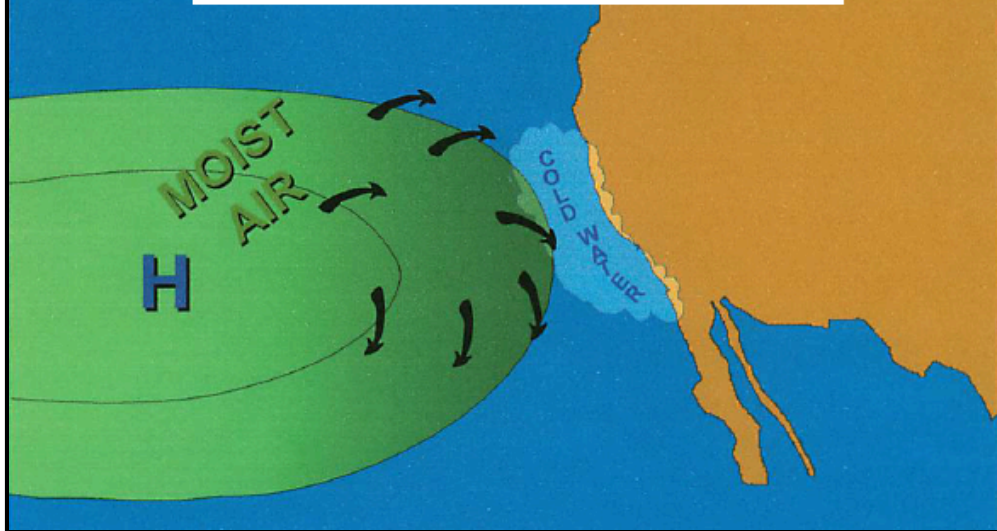
Advection fog will likely dissipate when the colder surface warms or when increasing turbulent mixing (i.e., winds increasing above 15 knots) lifts fog into a low cloud deck.

NOTE: When surface winds cease or change direction, fog often continues, but simply stops moving.



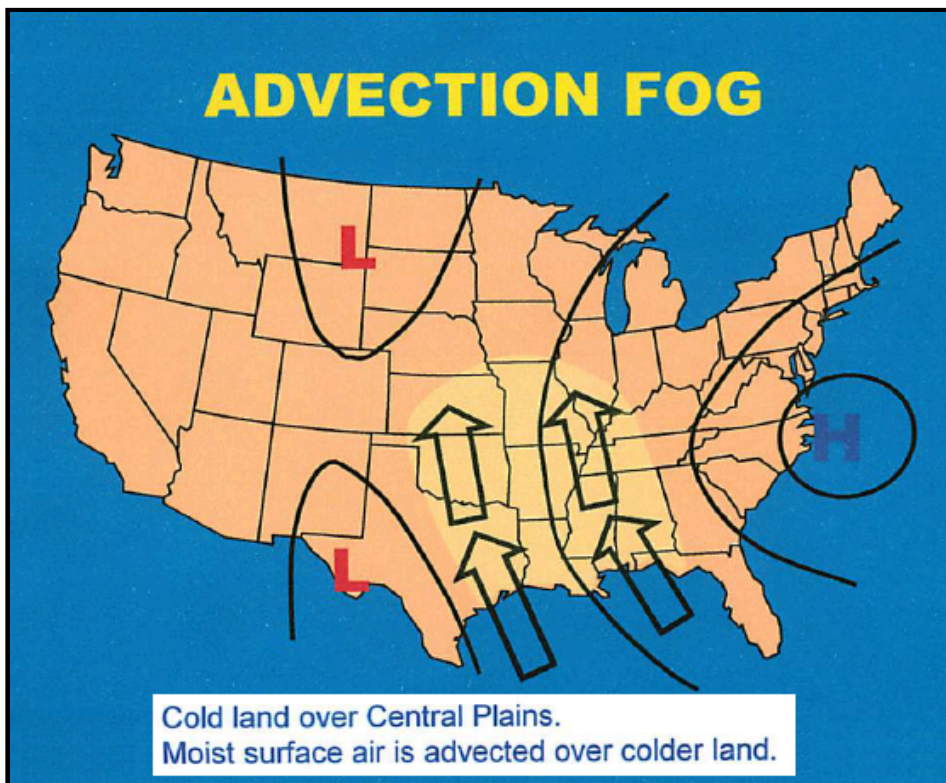
ADVECTION FOG

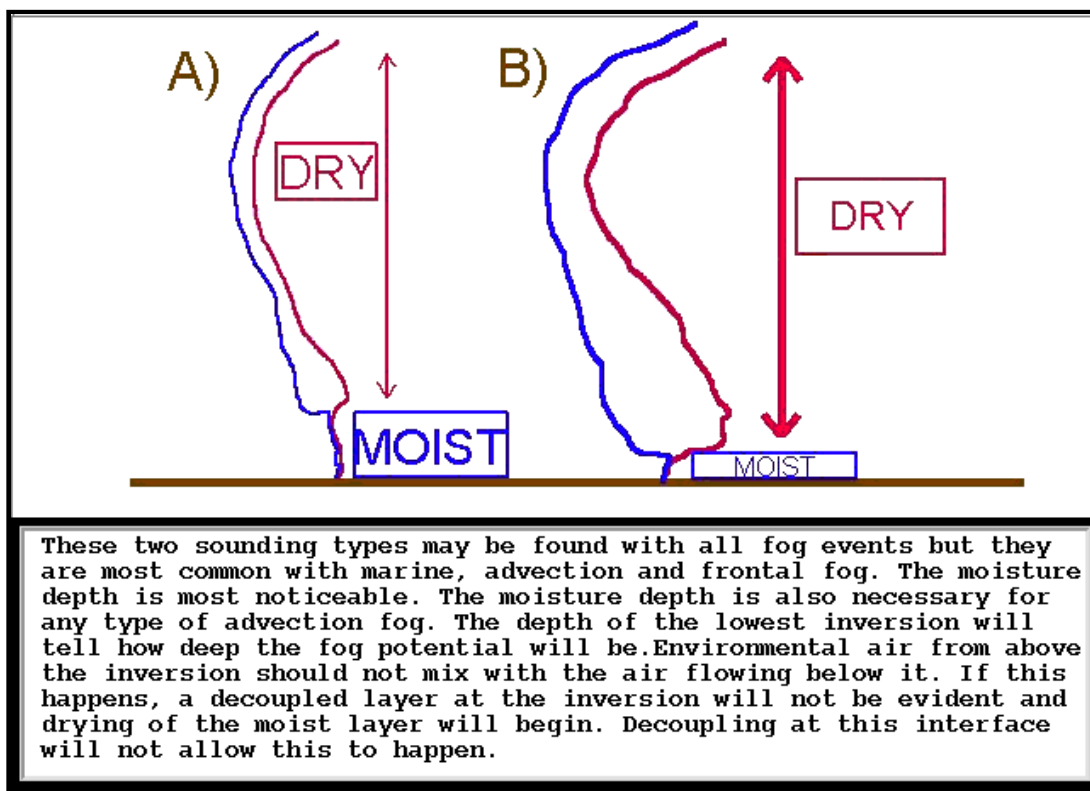
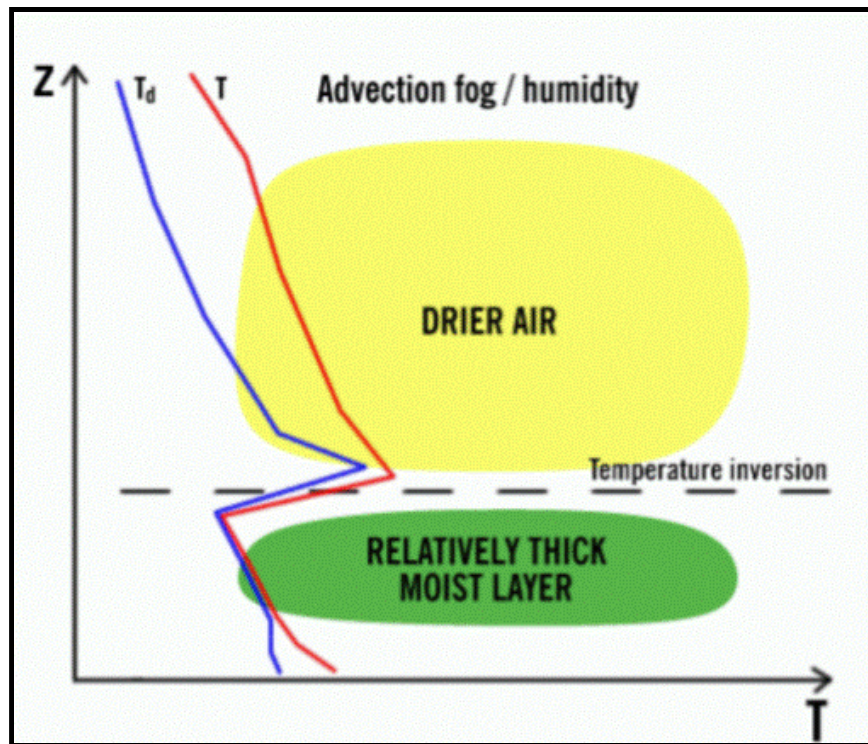
Colder surface water along the West Coast.
Moist surface air is advected over colder water.



ADVECTION FOG

Cold land over Central Plains.
Moist surface air is advected over colder land.





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